19.1-AI-ASS
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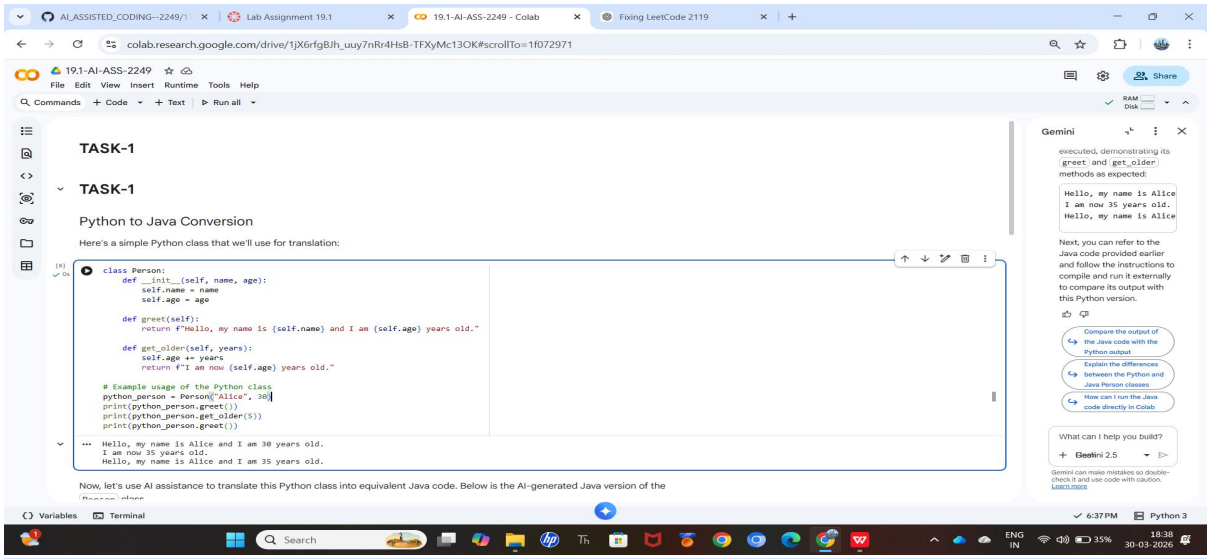
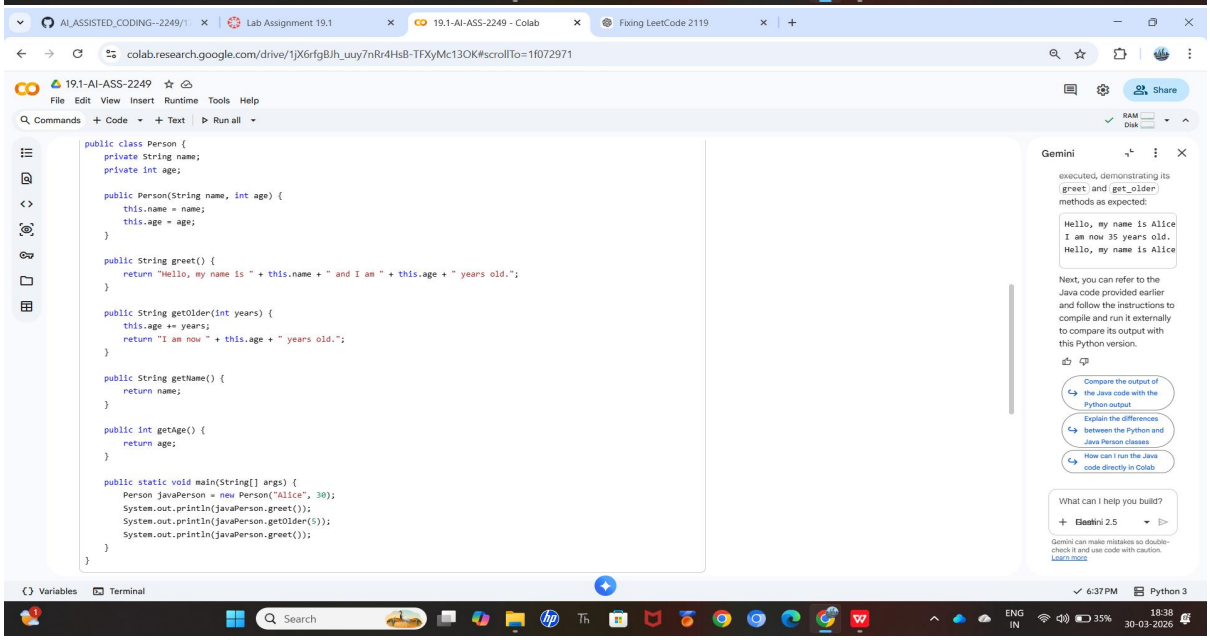
TASK-1

Prompt: Translate a simple Python class into Java using AI assistance.
Instructions:
Prompt AI to convert a Python class with attributes and methods into equivalent Java code.
Verify that constructors, method signatures, and data types are properly handled.
Run both versions (Python & Java) to compare expected outputs.
Expected Output: Equivalent working Java class translated from Python.

Code & Output:





Observation: Python to Java Conversion (Observation)

AI successfully translated Python class structures into Java by adapting dynamic typing into static typing. Constructors, data types, and method signatures required careful verification. Minor adjustments were needed for syntax differences, but functionality remained consistent across both implementations.

TASK-2

Prompt: Convert a C++ function for factorial calculation into Python.

Instructions:

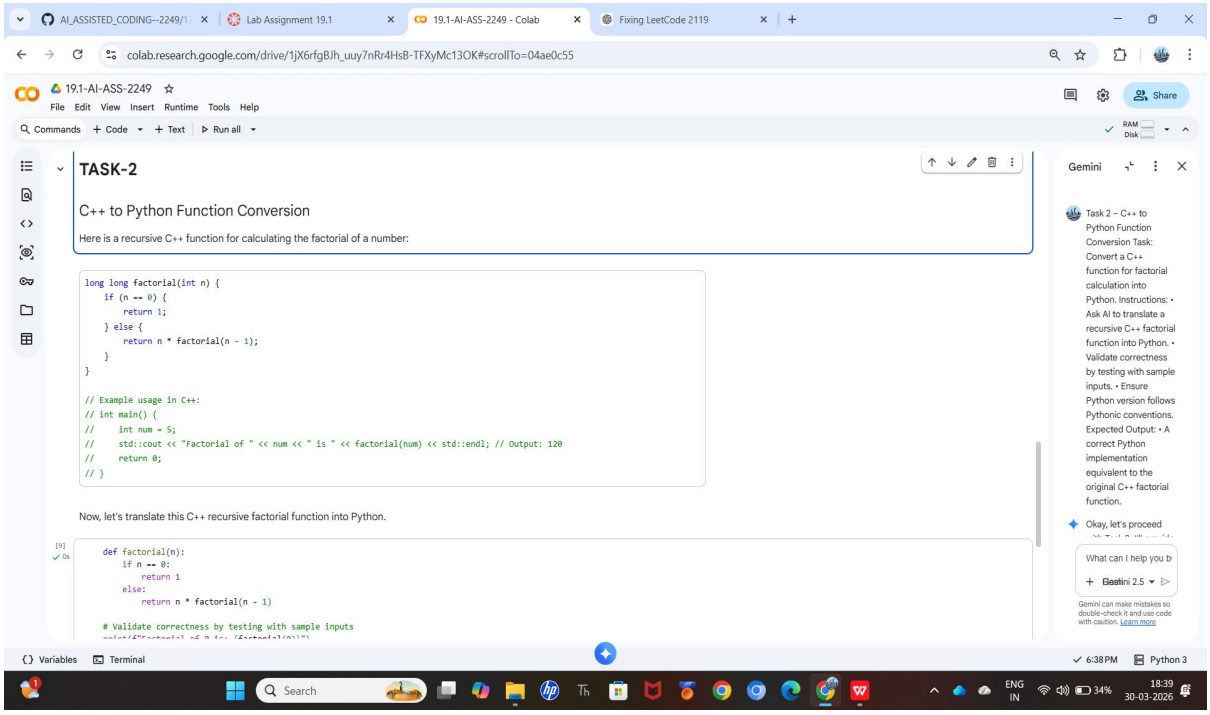
Ask AI to translate a recursive C++ factorial function into Python.

Validate correctness by testing with sample inputs.

Ensure Python version follows Pythonic conventions.

Expected Output: A correct Python implementation equivalent to the original C++ factorial function**.**

Code & Output:



Now, let's translate this C++ recursive factorial function into Python.

```
def factorial(n):
    if n == 0:
        return 1
    else:
        return n * factorial(n - 1)

# Validate correctness by testing with sample inputs
print(f"Factorial of 0 is: {factorial(0)}")
print(f"Factorial of 1 is: {factorial(1)}")
print(f"Factorial of 5 is: {factorial(5)}")
print(f"Factorial of 10 is: {factorial(10)}")
```

```
Factorial of 0 is: 1
Factorial of 1 is: 1
Factorial of 5 is: 120
Factorial of 10 is: 3628800
```

Observation: C++ to Python Function Conversion (Observation)

The recursive factorial function was accurately converted into Python with simpler and more readable syntax. Python's concise structure reduced code complexity. Validation with test cases confirmed equivalent outputs, demonstrating effective cross-language translation.

TASK-3

Prompt: Translate a SQL query into a Pandas DataFrame operation in Python.

Instructions:

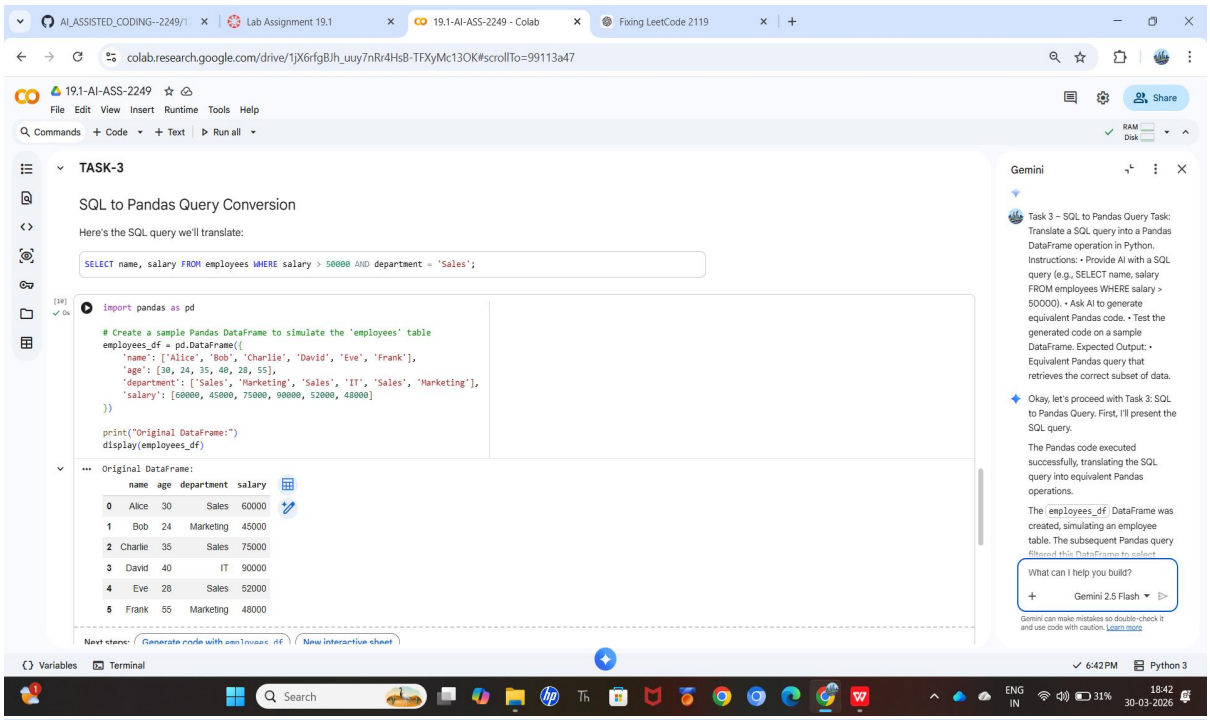
Provide AI with a SQL query (e.g., SELECT name, salary FROM employees WHERE salary > 50000).

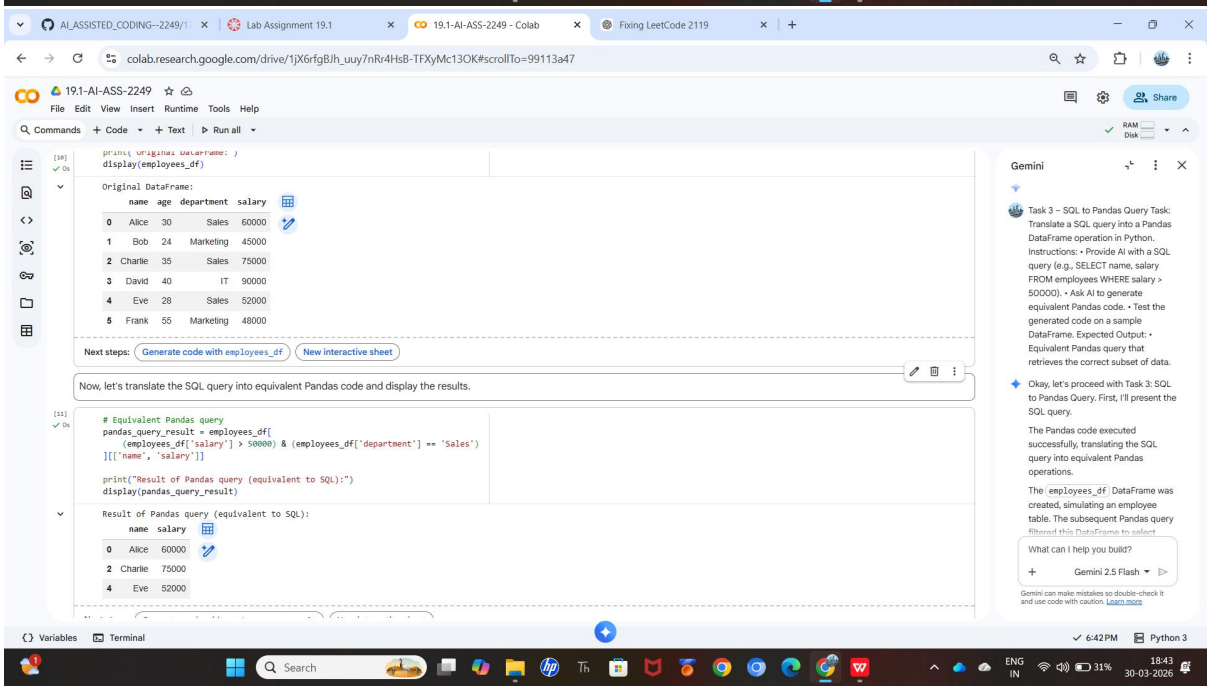
Ask AI to generate equivalent Pandas code.

Test the generated code on a sample DataFrame.

Expected Output: Equivalent Pandas query that retrieves the correct subset of data.

Code & Output:





Observation: SQL to Pandas Query (Observation)

AI effectively mapped SQL operations to Pandas DataFrame syntax. Filtering conditions and column selection were translated correctly using boolean indexing. The Pandas implementation produced the same results as the SQL query, ensuring data consistency.

TASK-4

Real-Time Application: Multi-Language Snippet Converter

Scenario: Students will choose a real-time use case where they need code in multiple languages.

Example: Converting a Python web scraping snippet into JavaScript for browser automation.

Translating a Java method into C# for enterprise applications.

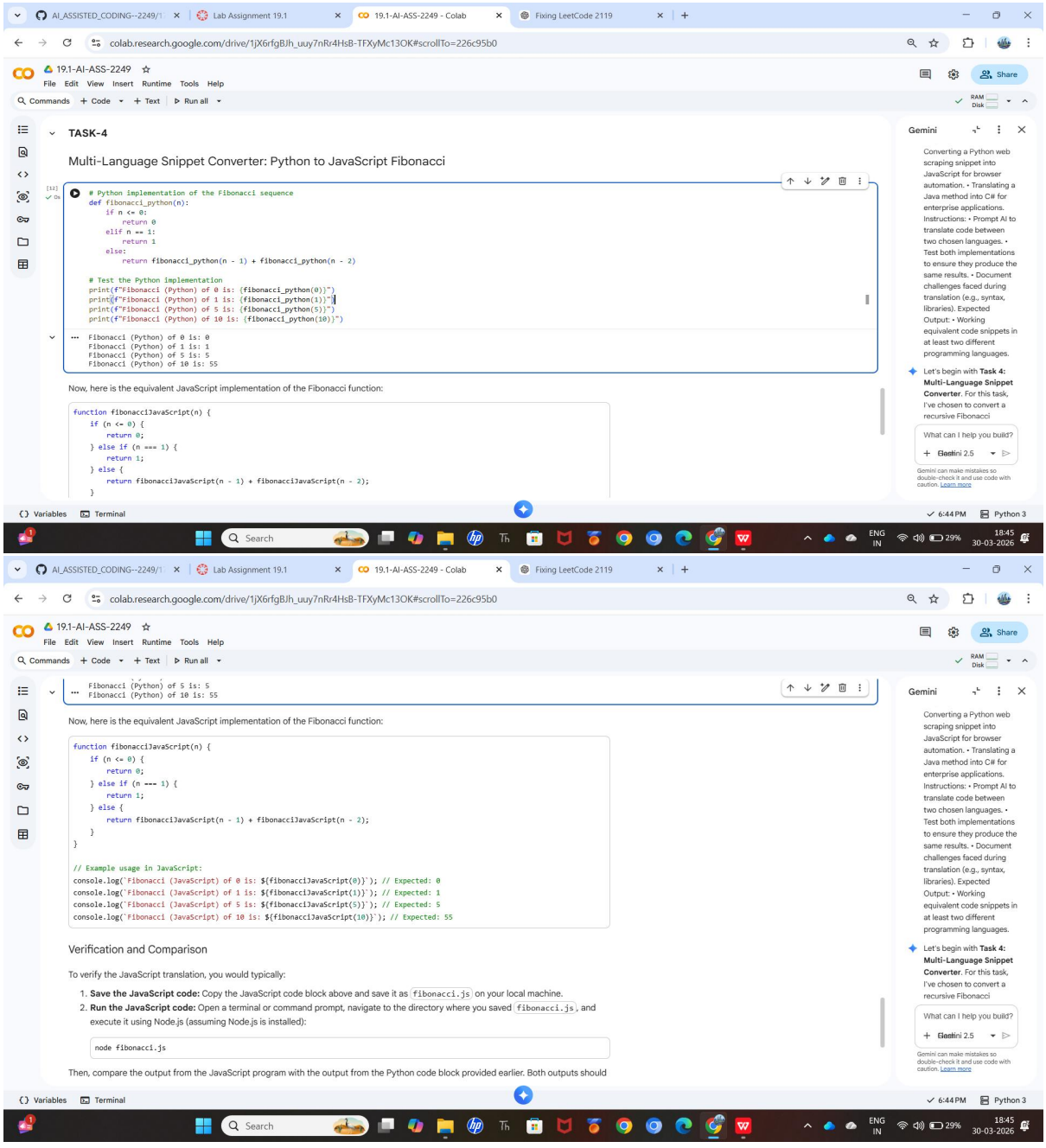
Prompt:

AI to translate code between two chosen languages.

Test both implementations to ensure they produce the same results.

Document challenges faced during translation (e.g., syntax,libraries).

Code & Output:



Observation: Multi-Language Snippet Converter (Observation)

Code translation between different languages was achieved with functional equivalence, but challenges arose due to differences in libraries, syntax, and runtime environments. Testing was essential to ensure accuracy. AI significantly reduced development effort while highlighting the need for manual optimization and debugging.